Marta Rutkowska-Kaine

- Durham based
- BSc Psychology
- 6 years of Administrative experience
- Passion for Fitness
- Data in daily life

Durham Cathedral

Sleep, Activity & Cognitive Performance Analysis

Exploring relationships between physical activity, sleep duration & efficiency, and cognitive performance

Two datasets were used (Kaggle):

- Fitbit Dataset
  - Daily activity, steps, calories, and sleep data
  - Real-world wearable device data
- Sleep Deprivation & Cognitive Performance Dataset
  - Reaction time, memory accuracy, and attention metrics
  - Includes sleep duration and psychological factors

These datasets were analysed separately to explore different aspects of sleep behaviour

Data Transformation

Cognitive Performance Dataset

- Checked for missing values and inconsistencies
- Verified data types and units of measurement
- Renamed columns for clarity
- Removed unnecessary fields

```
Column                      Non-Null Count
------                      --------------
Participant_ID              60 non-null
Sleep_Hours                 60 non-null
Sleep_Quality_Score         60 non-null
Daytime_Sleepiness          60 non-null
Stroop_Task_Reaction_Time   60 non-null
N_Back_Accuracy             60 non-null
Emotion_Regulation_Score    60 non-null
PVT_Reaction_Time           60 non-null
Age                         60 non-null
Gender                      60 non-null
BMI                         60 non-null
Caffeine_Intake             60 non-null
Physical_Activity_Level     60 non-null
Stress_Level                60 non-null
```

Data Transformation

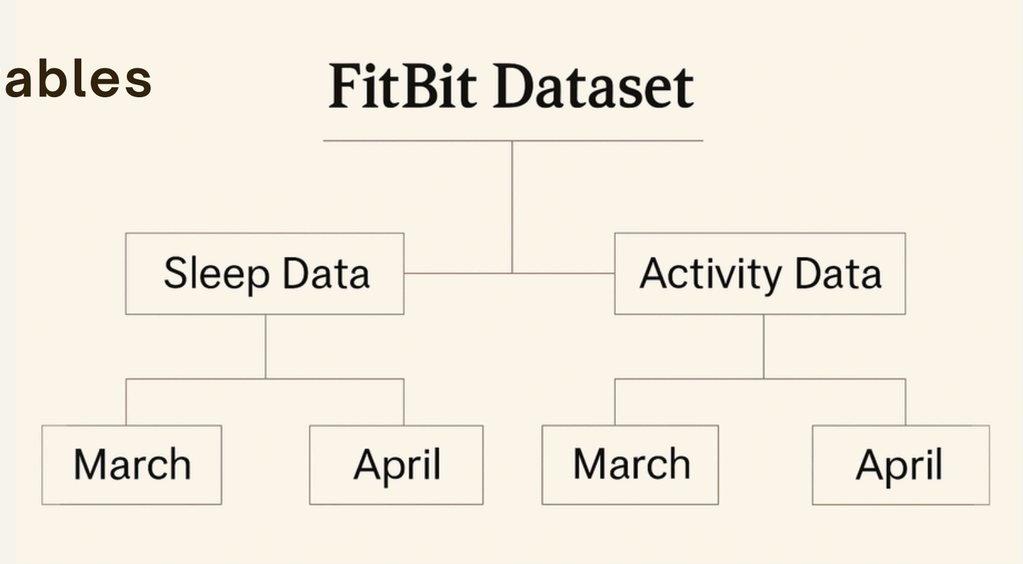
Calculating sleep hours and merging the tables

Adjusted timestamps by **12 hours** to correctly group overnight sleep into a single “sleep day

Aggregated minute-level data into **daily summaries**:

Converted sleep duration into **hours**

Created **sleep efficiency** metric (time asleep / time in bed)



sleep_day	minutes_asleep	minutes_restless	minutes_awake	sleep_hours	restless_hours	awake_hours	time_in_bed	sleep_efficiency
2016-03-12	411	15.0	0.0	6.850000	0.250000	0.000000	7.100000	0.964789
2016-03-13	354	27.0	5.0	5.900000	0.450000	0.083333	6.433333	0.917098
2016-03-14	312	16.0	7.0	5.200000	0.266667	0.116667	5.583333	0.931343
2016-03-15	272	26.0	5.0	4.533333	0.433333	0.083333	5.050000	0.897690
2016-03-16	463	36.0	1.0	7.716667	0.600000	0.016667	8.333333	0.926000

Data Transformation

Final merge and preparation for analysis

- Aggregated activity data that was separated into different records
- Standardised date formats to ensure accurate joins
- Merged sleep data with daily activity data using:
  - -User ID
  - -Date
- Created categorical sleep groups (low, moderate, high)
- Added weekday information for behavioural analysis (Weekday vs. Weekend)
- Removed outliers (sleep > 12 hours)

```
Column                 Non-Null Count
------                 --------------
Id                     562 non-null
ActivityDate           562 non-null
TotalSteps             562 non-null
Calories               562 non-null
VeryActiveMinutes      562 non-null
FairlyActiveMinutes    562 non-null
LightlyActiveMinutes   562 non-null
SedentaryMinutes       562 non-null
sleep_hours            562 non-null
minutes_asleep         562 non-null
minutes_restless       562 non-null
minutes_awake          562 non-null
time_in_bed            562 non-null
sleep_efficiency       562 non-null
weekday                562 non-null
sleep_category         562 non-null
```

Initial findings

Cognitive performance

- Subjects averaged **5.8 hours** of sleep, indicating sleep deprivation
- Lower sleep is associated with **slower reaction** times
- However, the relationship is **weak**, suggesting other factors also play a role
- Performance in the **PVT (Psychomotor Vigilance Test)** drops significantly with less than **5 hours** of sleep

	Sleep_Hours
count	60.000000
mean	5.805500
std	1.832357
min	3.120000
25%	4.107500
50%	5.690000
75%	7.285000
max	8.820000

<5 hours	353.985385
5-7 hours	314.017059
>7 hours	318.260588

Initial findings

Cognitive performance

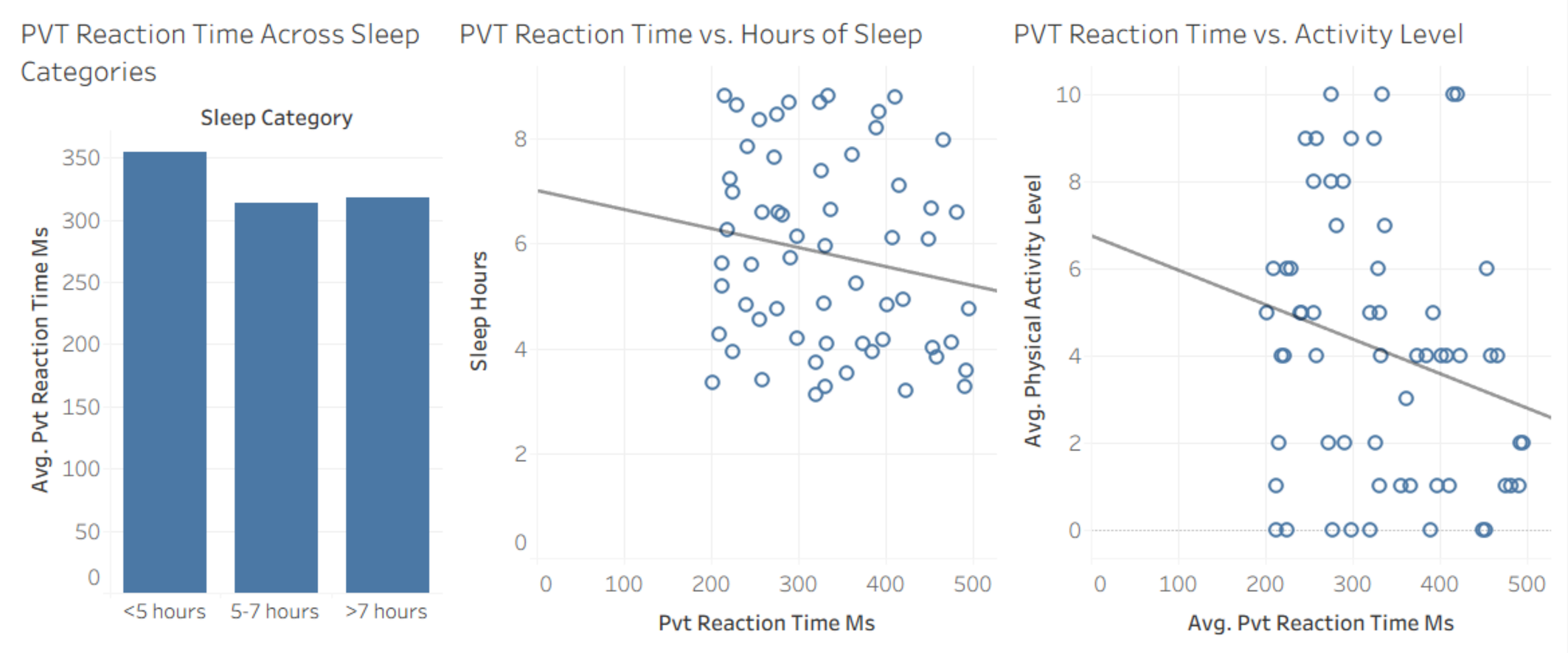
- **Physical activity** appears to have a stronger relationship with performance, especially **PVT**, than sleep alone
- Higher **BMI** is correlated with slightly higher sleep **quality**

	bmi
count	60.000000
mean	27.329333
std	4.541382
min	18.740000
25%	23.580000
50%	27.365000
75%	30.747500
max	34.930000

	pvt_reaction_time_ms	sleep_hours	sleep_quality_score	physical_activity_level	bmi
pvt_reaction_time_ms	1.000000	-0.174171	-0.131605	-0.224303	-0.148436
sleep_hours	-0.174171	1.000000	0.013907	0.153093	-0.034617
sleep_quality_score	-0.131605	0.013907	1.000000	-0.019553	0.203753
physical_activity_level	-0.224303	0.153093	-0.019553	1.000000	0.019635
bmi	-0.148436	-0.034617	0.203753	0.019635	1.000000

Below 18.5	Underweight
18.5 - 24.9	Normal Weight
25 - 29.9	Overweight
30 & Above	Obese

Visualisations Cognitive performance



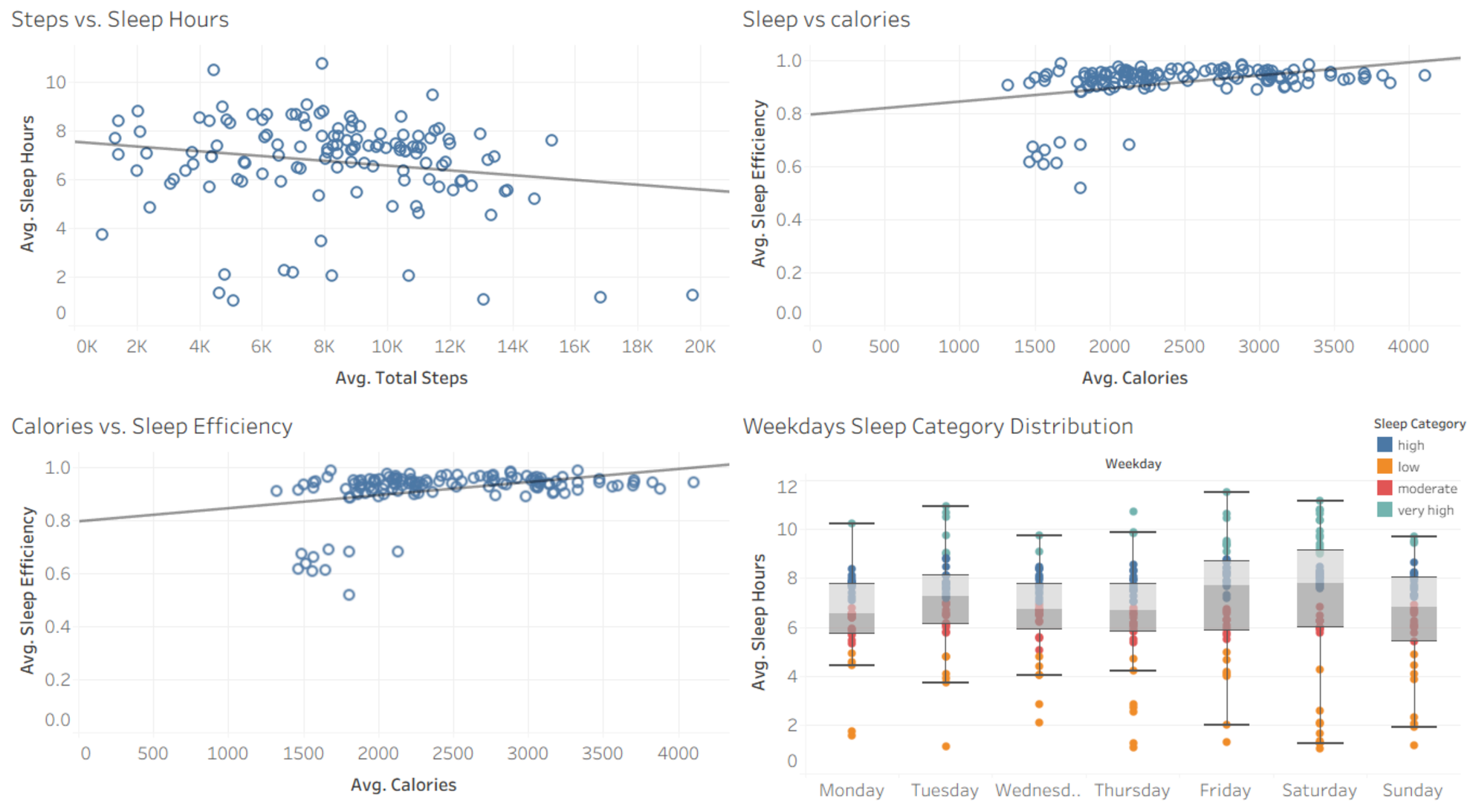
Initial findings

FitBit

- Physical activity shows weak relationships with sleep duration or quality
- Energy expenditure may have a slight positive effect on sleep efficiency
- Overall, sleep behaviour appears influenced by multiple factors

	sleep_hours	sleep_efficiency	TotalSteps	Calories	SedentaryMinutes	VeryActiveMinutes
sleep_hours	1.000000	0.271638	-0.095598	0.037359	-0.194534	-0.035387
sleep_efficiency	0.271638	1.000000	-0.094894	0.317434	0.074829	0.078831
TotalSteps	-0.095598	-0.094894	1.000000	0.328573	-0.273100	0.554924
Calories	0.037359	0.317434	0.328573	1.000000	-0.050558	0.586395
SedentaryMinutes	-0.194534	0.074829	-0.273100	-0.050558	1.000000	-0.107196
VeryActiveMinutes	-0.035387	0.078831	0.554924	0.586395	-0.107196	1.000000

Visualisations FitBit



Key Findings

- Sleep duration and quality show weak relationships with both activity and cognitive performance
- Sleep efficiency is more strongly linked to physical activity levels
- Calorie expenditure improves sleep duration and quality

The End

Thank You For Listening

LinkedIn

linkedin.com/in/marta-rutkowska-kaine/

GitHub

github.com/MartaKaine